

Network and VPN Setup Guide: Penang

Document ID: it-network-vpn-guide-penang. Site: Penang, Malaysia (factory, Bayan Lepas, site code PEN). Owner: IT Operations, Network team, with Penang OT Support. Last updated: 7 April 2025.

Who this guide is for

New starters and visitors working at the Penang factory offices, and Penang-based staff connecting from home. Other sites have their own guide; subnets, VPN endpoints and printer queues are different at each site. Penang is the only site with a separate operational technology (OT) network for production.

Wi-Fi

- Meridian-Corp: company laptops only, 802.1X certificate authentication. Not available on the production floor.
- Meridian-Guest: visitors, sponsor approval by email, sessions last 8 hours.
- There is no Wi-Fi on the OT network. Shop-floor kiosks are wired.

Site subnets

Network | VLAN | Subnet

Corporate supernet | n/a | 10.40.0.0/16

Users (wired and Meridian-Corp) | 410 | 10.40.16.0/20

Voice | 420 | 10.40.32.0/23

Printers | 430 | 10.40.40.0/24

Site servers | 440 | 10.40.100.0/24

OT network (SMT lines, test stations, calibration stations) | separate | 10.41.0.0/16

DNS servers: 10.40.1.10 and 10.40.1.11. Internal domain: meridian.internal. The OT network 10.41.0.0/16 has its own DNS and cannot be reached directly from the corporate network or the VPN.

VPN (Meridian Connect)

- Client: Meridian Connect, installed on every company laptop.
- Penang endpoint: vpn-pen.meridian-instruments.com
- Protocol and port: UDP 443. Penang uses UDP 443 instead of the UDP 51820 used at other sites because several local home broadband providers filter non-standard UDP ports.
- Tunnel mode: split tunnel. Traffic to 10.0.0.0/8 and meridian.internal goes through the VPN.
- Authentication: directory account plus MFA.
- Idle timeout: 8 hours.

If vpn-pen is unavailable, use vpn-lds.meridian-instruments.com as the fallback (note it listens on UDP 51820).

Remote access to the OT network

Access to the OT network from the VPN is only through the jump host ot-jump-pen-01.meridian.internal. It requires a FIDO2 security key, an approved Beacon ticket in category MFG-OT, and a session recorded by the privileged access vault. Sessions are limited to 4 hours.

Printers

Print server: prn-pen-01.meridian.internal. Queues:

Queue | Location | Type

PEN-ADM-MFP01 | Administration building | Colour multifunction

PEN-SMT-LABEL01 to PEN-SMT-LABEL04 | One per SMT line (SMT-1 to SMT-4) | Board serial label printers

PEN-FT-LABEL01 | Final test and calibration | Product and calibration labels

PEN-WH-ZEBRA01 | Warehouse | Shipping labels

Label printers on the SMT lines are on the OT network and are managed by Penang OT Support, not the corporate print server.

Getting help

Raise a Beacon ticket in category NET-VPN, NET-WIFI or PRN-QUEUE for corporate issues, and MFG-OT for anything on the production floor. The Penang IT desk is in the administration building, open 07:30 to 17:30 MYT.